

Cellular telephony

4.1 Introduction and history

Mobile phones, otherwise known as cellular phones or cell phones, are possibly the best things since sliced bread. While all mobile phones allow quick communication on the move from a lightweight, low power package, some of the newer features also include high speed internet access, multimedia messages, etc.

Let us take a brief look at how this wonderful technology came into being.

The origin of mobile phones can be found in the early days of the development of radio.

Initially, it was just two way radios. While these radios could call each other, they could not call a traditional fixed line phone. These types of radios were bulky, often weighing tens of kilograms and were intended to be used only as car phones. They gained a large customer base amongst truck drivers and other travellers.



A modern cell phone

Soon, attempts were made to overcome the limitations of these designs. In 1945, an experimental system was tested wherein a single base station (similar to a mobile phone tower of today) transmitting a powerful signal covered a whole area. It could operate just one phone at the same time, as concepts such as sharing frequencies were not defined until the 1970s. It was similar to the cordless telephone of today. This technology is now known as 0G (zero generation) and saw no commercial applications.

The inventor of the first practical mobile phone that could be used in a non-vehicle environment

From car phones to GSM, CDMA and WCDMA to even satellite phones, wireless communication has grown significantly over the years